

DDTA research work plan after documentation-layer closure

WORK PLAN - REVISION 7

Status: active BA1 execution plan after BA1-T2 split-or-collapse pressure testing.

Supersedes: Revision 6 only for forward execution state; R1-R6 remain historical research records.

1. Fixed prior state

- Chapters 2-4: CLOSED / FINAL for current thesis scope.
- Documentation layer: CLOSED.
- BA0-R systems-modeling prior-art research: CLOSED.
- BA0 responsibility/non-goals: CLOSED by BA0-T3.
- ThreatForge is a case study/tooling instrument, not DDTA semantic authority.
- No historical focus set, systems-modeling metaclass or threat-method element is inherited automatically into BA1.

2. BA0 boundary carried forward

Base Analysis must preserve baseline-scoped shared semantic identity, source/origin provenance and explicit unresolved state sufficiently for reusable human/method projections, source drill-down, change-impact reasoning and source-localized feedback handoff.

The representation remains methodology-neutral and implementation-independent. Grounded, derived and diagnostic/unresolved are responsibility-level origin/state distinctions; a generic reviewed analytical-addition class is not required.

3. BA1 objective

BA1 decides the **minimal first-class semantic identity ontology** needed to realize the BA0 boundary.

A first-class split requires both:

1. independent semantic identity across propositions/projections/change; and
2. reusable subtype-specific invariants that cannot be represented honestly as classifications, roles, values or propositions over an existing family.

```
semantic responsibility exists
    !=
first-class metaclass required
```

4. BA1-T1 - minimal ontology candidate derivation

Status: COMPLETED / PROVISIONAL CANDIDATE.

BA1-T1 produced the initial falsifiable candidate:

```
BAReferent
BAProposition
```

It rejected a fully undifferentiated analytical element and did not force domain-specific roots.

5. BA1-T2 - split-or-collapse pressure test

Status: COMPLETED / PROVISIONAL PASS WITH MINIMALITY REFINEMENT.

BA1-T2 deliberately stressed the two-family candidate against:

- facial-access delivery behavior and specialized constraints;

- order-fulfillment physical handoff reuse;
- lifecycle/correlation-heavy informational meanings;
- internal/external responsibility placement;
- store and contract candidates;
- state/indeterminate semantics;
- a bounded STRIDE-oriented method consumer;
- semantic collapse into one undifferentiated family.

Result

No tested semantic kind forced a dedicated first-class split.

Behavior/Event	NOT FORCED
Information/Resource	NOT FORCED
Participant/Component/Capability	NOT FORCED
Boundary/Externality	NOT FORCED
Store	NOT FORCED
Contract	NOT FORCED
State/Mode/Context	NOT FORCED

The semantic one-family collapse was **REJECTED** because referent identity and proposition identity/origin/evolution remain independently required.

Minimality refinement

R1 allowed **BAProposition** to be a project-semantic target of another proposition. T2 found this unnecessary and potentially misleading.

The refined boundary is:

- project meaning that needs independent reuse/qualification -> **BAReferent**;
- methodology-neutral assertion about project meaning -> **BAProposition**;
- assertion provenance/diagnostic/change metadata -> BA3 mechanisms targeting proposition identity.

The active candidate is now **BA1_MINIMAL_BAE_ONTOLOGY_CANDIDATE_R2.md**.

6. BA1-T3 - minimal BAE ontology closure review

NEXT / NOT EXECUTED BY THIS PLAN REVISION.

BA1-T3 is a closure/falsification review, not another ontology invention exercise.

It must test:

1. **Necessity:** does removing either **BAReferent** or **BAProposition** break a closed BA0 responsibility or observed corpus requirement?
2. **Sufficiency:** does the two-family boundary cover the tested corpora without hiding a still-unresolved first-class identity responsibility?
3. **No leakage:** can the boundary be stated without relation vocabulary, document types, STRIDE/DFD categories, ThreatForge classes, view/tool/storage semantics or general systems-modeling taxonomies?
4. **Correct phase placement:** do predicates, roles and actions belong to BA2; provenance and identity mechanics to BA3; projections to BA4; vocabulary assistance to BA5; and regression to BA6?
5. **Minimality refinement:** is proposition-as-project-target correctly excluded from the required core?
6. **Stop criterion:** if no material counterexample remains and unresolved details belong later, close BA1 rather than invent BA1-T4 for reassurance.

Possible exit

If the review passes:

BA1 CLOSED
BAReferent ACCEPTED first-class identity family
BAProposition ACCEPTED first-class identity family
BA2 NEXT

If a material counterexample remains, record it explicitly and keep BA1 open. Do not silently push unresolved ontology into BA2.

7. Sequence after BA1 closure

Do not skip phases:

BA2 - Relations and canonical action vocabulary
BA3 - Document-to-BAE derivation, provenance and authority
BA4 - Multi-level human and method projections
BA5 - Controlled vocabulary and optional authoring/extraction assistance
BA6 - Base Analysis closure and regression

BA2

Define the smallest stable relation/action vocabulary over the accepted BA1 identity ontology. Keep semantic relation, canonical lexical label and authoring synonym assistance separate.

BA2 may determine proposition arity, participation roles, dependency/constraint/action semantics and how reusable semantic classifications are represented, but it must not retroactively create BA1 root types without a concrete reopen trigger.

BA3

Close identity/equivalence/lifecycle rules, source locators, grounded/derived/diagnostic materialization, acceptance/rejection, stale state and document-to-analysis traceability.

BA4

Define derived human and method projections. No view is source authority.

BA5

Define controlled vocabulary and optional assistance. LLM/NLP proposals remain candidate-producing support and cannot establish authority silently.

BA6

Regress the Base Analysis against closed corpora and at least one structurally different holdout before formal overlay work.

8. Analysis envelope remains downstream

Only after Base Analysis closure:

- A1 - AnalysisRecord / execution envelope;
- A2 - common reviewed Finding boundary;
- A3 - change/provenance integration;
- formal methodology overlays and STRIDE/STRIDE-AI evaluations.

The BA0-T2 STRIDE probe remains only a pre-overlay consumer probe.

9. Next authorized microstep

Execute only:

BA1-T3 - minimal BAE ontology closure review.

Do not start BA2, formal STRIDE overlay design, Common Finding schema or implementation work until BA1 closes explicitly.